

BUFORD PERTINENT DATA

GENERAL

Location (damsite) – Gwinnett and Forsyth Counties, GA, miles above mouth of Chattahoochee River	348.3
Drainage area above damsite - square miles	1,034

RESERVOIR

Length at elevation 1,070 feet NGVD29 - river miles Chattahoochee River/Chestatee River	44/19
Top of conservation pool elevation, summer/winter - feet NGVD29	1,071/1,070
Peak pool for standard project flood – feet NGVD29	1,085
Peak pool for spillway design flood – feet NGVD29	1,100
Top of flood risk management pool - feet NGVD29	1,085
Bottom of conservation pool elevation – feet NGVD29	1,035
Area at top of conservation (elevation 1,071) – acres	38,425
Conservation storage elevation 1,071 – 1,035 – acre feet	1,074,645
Length of shoreline (elevation 1,071) – miles	692

TAILWATER ELEVATIONS

Normal, service unit only – feet NGVD29	912.2
Normal, one large unit and service unit operating (outflow 6,000 cfs) – feet NGVD29	917.0
Normal, 3 units operating (outflow 11,200 cfs) – feet NGVD29	920.3

DAM/EARTH DIKES

Dam total length - feet	1,630
top elevation - feet NGVD29	1,106
Saddle Dikes (1, 2, and 3), total length - feet	6,600
top elevation - feet NGVD29	1,106

EMERGENCY SPILLWAY

Type	Uncontrolled chute
Total Width - feet	100
Crest elevation – feet NGVD29	1,085

FLOOD RISK MANAGEMENT SLUICE

Number of sluices / Number of gates (each gate has jet valve)	1 / 2
Discharge capacity of jet valve – cfs	600
Discharge capacity at elevation 1,085 / 1,070 NGVD29 - cfs	11,590 / 11,030

POWER PLANT

Generating capacity (declared*) - MW (2 units @ 60, 1 unit @ 7)	127
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* Declared generating capacity is defined as the plant's operational capacity declared on a weekly basis to the power marketing agency. The value may vary slightly from week to week depending on factors such as head and cooling capabilities; values shown are the nominal values reported.

STREAM FLOW

Drainage area at dam site-square miles	1,034
Minimum mean monthly flow before construction (1903–1956) (September 1925)-cfs	264
Estimated minimum daily flow before construction (1903–1956) (August 1925)-cfs	119
Average annual flow Before construction (1903–1956)-cfs	2,042
Maximum mean monthly flow before construction (December 1932)-cfs	8,642
Maximum recorded flow before construction (January 8, 1946)-cfs	55,700
Minimum mean monthly and daily flows after construction can only be estimated because of measuring techniques. Unimpaired flows were developed to estimate flows with no reservoir (and other) effects.	
Average annual flow after construction (1957–2013) measured at the dam-cfs	1,914
Discharge at bankfull stage-cfs	10,000
With reservoir in place, peak flow estimated (September 9, 2009)-cfs	71,700
Minimum mean monthly flow after construction (1956–2011) based on unimpaired flows occurred July 1986-cfs	195
Average annual flow after construction (1956–2011) based on unimpaired flows-cfs	2,034

SPILLWAY-DESIGN FLOOD

National Weather Service 72-hour storm at Longitude 85°47' Latitude 35°34'	
Total rainfall-inches	30.67
Initial loss-inches	0.00
Average infiltration rate-inches per hour	0.04
Total storm runoff-inches	28.52
Total volume of storm runoff-acre feet	1,581,600
Peak rates of flow	
Inflow to full reservoir-cfs	581,300
Total reservoir outflow-cfs	40,400
Spillway discharge-cfs	28,400
Duration of flood-days	5

RESERVOIR

Pool elevations-(feet NGVD29)

Maximum pool, spillway design flood	1,100
Top of flood risk management pool	1,085
Top of conservation pool	
Summer	1,071
Winter	1,070
Minimum conservation pool	1,035
Streambed (bottom of flood risk management sluice)	919

Storage volumes-(acre feet)

Maximum pool, spillway design flood	3,332,000
Total storage-elev 1,085	2,551,064
Flood risk management storage, elev 1,085 to 1,070 (11.48")	640,264
Flood risk management storage, elev 1,085 to 1,071 (10.80")	598,800
Conservation storage, elev 1,071 to 1,035 (19.61")	1,074,645
Conservation storage, elev 1,070 to 1,035 (18.92")	1,036,532
Inactive storage, below elev 1,035	874,268

Reservoir areas-(acres)

Maximum pool, spillway design flood, elev 1,100	57,601
Top of flood risk management pool, elev 1,085	48,176
Top of conservation pool, elev 1,071 / 1,070	38,425 / 37,871
Top of inactive storage, elev 1,035	22,293
Area (within taking line elev 1,085 plus small islands/peninsulas)-acres	
Purchased in fee simple	56,155
Right to inundate acquired by easement	719
River bed	1,133
Total	58,007

Length of shoreline-(miles)

Top of flood risk management pool, elev 1,085	760
Top of conservation pool, elev 1,070	540

Length of reservoir at elevation 1,070-(river miles)

Chattahoochee River	44
Chestatee River	19

DAM

Type	Rolled-fill earth
Length along crest of main dam-feet	1,630
Top width-feet	40
Base width-feet	1,000
Height of main dam above river bed-feet	192
Total length of saddle dikes-feet	6,600
Elevation, top of dam and saddle dikes-feet NGVD29	1,106

SPILLWAY

Type	Uncontrolled chute
Width of chute-feet	100
Crest elevation-feet NGVD29	1,085

FLOOD RISK MANAGEMENT SLUICE

Number of sluices	1
Number of gates	2
Type of gates	Broome
Size of gate	6.5 x 13.25
Number of jet valves per gate	1
Size of jet valve-inches diameter	36
Discharge capacity through jet valve-cfs each	600
Discharge capacity at elev. 1,085-cfs	11,590
Discharge capacity at elev. 1,070-cfs	11,030
Discharge capacity at elev. 1,035-cfs	10,080
Discharge capacity at elev. 919 (Invert)-cfs	0

POWERHOUSE

Size of building	
Length-feet	205
Width-feet	94.5
Type-Indoor, reinforced concrete and structural steel const.	
Elevation-feet NGVD29	
Bottom of substructure	885
Low point of draft tube	888
Centerline of distributor, 60,000 kw units	927
Centerline of distributor, 7,000 kw units	922.5
Generating units-kw	
One 7,000 kw unit	
Speed rpm	277
Turbines	
Type	Francis
Rotation	counter-clockwise
Guaranteed capacity at best gate, and 157.4 foot net head-hp	9,700
Two 60,000-kw units	
Spacing, center to center-feet	62
Speed-rpm	100
Turbines	
Type	Francis
Rotation	Clockwise
Guaranteed capacity at best gate, and 136-foot net head-hp (no air inflow)	79,500
Guaranteed capacity at best gate, and 136-foot net head-hp (with air inflow)	73,140
Generators	
One service unit	
Rated capacity, continuous kVA	7,870
Power factor	0.90
Voltage	13,800
Two large units each	
Rated capacity, continuous kVA	69,333
Power factor	0.90
Voltage	13,800

POWER DATA

Drawdown for storage-feet	35
Volume in power storage (elev. 1,035–1,070)-acre-feet	1,036,532
Rated net head, feet (2 Main Units)	148.4
Rated net head, feet (1 Small Unit)	157.4
Observed Tailwater elevations, feet NGVD29	
Normal, 3 units operating-outflow 11,200 cfs	920.3
Normal, 1 large unit and service unit 6,000	917.0
Service unit only	912.2
Plant output	
Average annual energy produced (1960–2013)-mwh	182,202